

Linear algebra: Exercise session 1

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1. For the following vectors:

$$v = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}, \quad w = \begin{bmatrix} 0 \\ 2 \\ 2 \end{bmatrix}.$$

- (a) Compute $3v$ (scalar multiplication) and $3v - w$ (linear combination);

Answer.

$$3v = 3 \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \\ -3 \end{bmatrix}, \quad 3v - w = \begin{bmatrix} 3 \\ 6 \\ -3 \end{bmatrix} - \begin{bmatrix} 0 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \\ -5 \end{bmatrix}.$$

- (b) Compute the length of w and v and check the triangle and the Cauchy Schwartz inequalities on v and w ;

Answer.

$$\|v\| = \sqrt{1^2 + 2^2 + (-1)^2} = \sqrt{6}, \quad \|w\| = \sqrt{0^2 + 2^2 + 2^2} = \sqrt{8} = 2\sqrt{2}.$$

Triangle inequality:

$$v + w = \begin{bmatrix} 1 \\ 4 \\ 1 \end{bmatrix}, \quad \|v + w\| = \sqrt{1 + 16 + 1} = \sqrt{18} = 3\sqrt{2},$$

$$\|v + w\| = 3\sqrt{2} \leq \sqrt{6} + 2\sqrt{2} = \|v\| + \|w\|.$$

Cauchy–Schwarz inequality:

$$\langle v, w \rangle = 1 \cdot 0 + 2 \cdot 2 + (-1) \cdot 2 = 2,$$

$$|\langle v, w \rangle| = 2 \leq 4\sqrt{3} = \|v\|\|w\|.$$

- (c) Compute $\|v\|^2 - \|w\|^2 + \|4v - 2w\| - \langle v + w, v - w \rangle$.

Answer.

First solution:

$$\|v\|^2 = 6, \quad \|w\|^2 = 8, \quad 4v - 2w = \begin{bmatrix} 4 \\ 4 \\ -8 \end{bmatrix}, \quad v + w = \begin{bmatrix} 1 \\ 4 \\ 1 \end{bmatrix}, \quad v - w = \begin{bmatrix} 1 \\ 0 \\ -3 \end{bmatrix},$$

$$\|4v - 2w\| = \sqrt{16 + 16 + 64} = \sqrt{96} = 4\sqrt{6}, \quad \langle v + w, v - w \rangle = 1 - 3 = -2.$$

Therefore,

$$\|v\|^2 - \|w\|^2 + \|4v - 2w\| - \langle v + w, v - w \rangle = 6 - 8 + 4\sqrt{6} - (-2) = 4\sqrt{6}.$$

Second solution:

$$\langle v + w, v - w \rangle = \langle v, v \rangle - \langle v, w \rangle + \langle w, v \rangle - \langle w, w \rangle = \|v\|^2 - \|w\|^2.$$

Thus,

$$\|v\|^2 - \|w\|^2 + \|4v - 2w\| - \langle v + w, v - w \rangle = \|4v - 2w\| = \sqrt{16 + 16 + 64} = \sqrt{96} = 4\sqrt{6}.$$

2. Are vectors $a = (1, -1)$ and $b = (2, 0)$ linearly independent? Why? Are they a basis for $\mathbb{R}^2$?

Answer.

Suppose $\alpha a + \beta b = (0, 0)$. That means

$$\alpha(1, -1) + \beta(2, 0) = (\alpha + 2\beta, -\alpha) = (0, 0),$$

which gives $\alpha = 0$, $\beta = 0$. Thus, a, b are linearly independent. Since there are two independent vectors in $\mathbb{R}^2$, they form a basis.

3. Are vectors $a = (1, 3, 0)$ and $b = (1, 0, 2)$ linearly independent? Why? Are they generating $\mathbb{R}^3$?

Answer.

Suppose $\alpha a + \beta b = (0, 0, 0)$. Then,

$$(\alpha + \beta, 3\alpha, 2\beta) = (0, 0, 0),$$

implying $\alpha = 0, \beta = 0$. Hence, a, b are linearly independent. But with only two vectors we cannot generate all of $\mathbb{R}^3$.

4. Are vectors $a = (1, 3, 0)$, $b = (1, 0, 2)$, and $c = (3, 3, 4)$ linearly independent? Why? Are they a basis for $\mathbb{R}^3$?

Answer.

$$a + 2b = (1 + 2, 3 + 0, 0 + 4) = (3, 3, 4) = c.$$

This means c is a combination of a, b . Thus, the three vectors are dependent and not a basis for $\mathbb{R}^3$.

5. If vector a is perpendicular to vectors b and c . Is it also perpendicular to $b + 2c$? Are b and c linearly dependent?

Answer.

$$a \cdot (b + 2c) = a \cdot b + 2(a \cdot c) = 0 + 2 \cdot 0 = 0.$$

Thus, a is also perpendicular to $b + 2c$. But nothing here shows whether b, c are dependent or independent. For example, it is possible that $b = -c$.

6. If a and b are perpendicular unit vectors. Show that $\|a - b\|^2 = 2$.

Answer.

$$\|a - b\|^2 = (a - b) \cdot (a - b) = \|a\|^2 + \|b\|^2 - 2(a \cdot b) = 1 + 1 - 0 = 2.$$

7. Compute the angle value between $a = (1, 0, -\sqrt{3})$ and $b = (3, 0, 0)$.

Answer.

$$\langle a, b \rangle = 3, \quad \|a\| = \sqrt{1 + 3} = 2, \quad \|b\| = 3.$$

$$\cos \theta = \frac{a \cdot b}{\|a\| \|b\|} = \frac{3}{2 \cdot 3} = \frac{1}{2},$$

Thus, $\theta = \pi/3 = 60^\circ$.

8. Write a MATLAB code that prompts the user to enter a scalar as the dimension of a vector space and displays the zeros, ones, and elementary vectors in this space.

Answer.

```
1  clc;
2
3  %Prompt the user to enter a number as the dimension of vector space
4
5  n = input('Enter a number as the dimension of vector space (e.g. 1, 2, ...), n = ');
6
7  %-----
8  %-----Zeros Vector-----
9
10 fprintf('0_%d = \n', n);
11 disp(zeros(n,1));
12
13 %-----
14 %-----Ones Vector-----
15
16 fprintf('1_%d = \n', n);
17 disp(ones(n, 1));
18
19 %-----
20 %-----Elementary Vectors-----
21
22 elementary = eye(n,n);
23
24 for i = 1: n
```

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25     fprintf('e_%d = \n',i);
26     disp(elementary(:,i));
27 end

```

9. Write a MATLAB code that prompts the user to enter a vector and a scalar, then displays the vector, the dimension of vector space containing the vector, the length (norm) of the vector, and the scalar multiplication of the scalar with the vector.

Answer.

```

1  clc;
2
3  %Prompt the user to enter a vector and a scalar
4
5  user_vector = input('Enter a vector (e.g. [1 2 3]): v = ');
6  user_scalar = input('Enter a Scalar      = ');
7
8  %-----
9  %-----Print Vector-----
10
11 fprintf('v = ');
12 disp(user_vector);
13
14 %-----
15 %-----Dimension of Space-----
16
17 %Notice that in MATLAB length of a vector is defined by the number of its elements, or
18 %dimension of vector space
19 %Initialize the length_vector to zero
20
21 length_vector = 0;
22
23 %Loop through each elements of the vector to count the elements
24
25 for i = user_vector
26     length_vector = length_vector +1;
27 end
28
29 %%% Another way
30 %%% length_vector = length(user_vector)
31
32 fprintf('The dimension of vector space is: %d\n', length_vector);
33
34 %-----
35 %-----Norm of Vector-----
36
37 %Initialize the norm to zero
38
39 norm_vector = 0;
40
41 %Compute the norm of vector
42
43 for i=1 : length_vector
44     norm_vector = norm_vector + user_vector (i)^2;
45 end
46
47 norm_vector = sqrt(norm_vector);
48
49 %%% Another ways
50 %%% 1) norm_vector = sqrt(sum(user_vector.^2));
51 %%% 2) norm_vector = norm(user_vector);
52
53 fprintf('||v|| = %.2f\n', norm_vector);
54
55 %-----
56 %-----Scalar Multiplication-----
57

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58 %Initialize the scalar multiplication vector to zero
59
60 s_multi = zeros(1,length_vector);
61
62 %Compute the norm of vector
63
64 for i=1 : length_vector
65     s_multi(i) = user_scalar * user_vector (i);
66 end
67
68 %%% Another way
69 %%% s_multi = user_scalar * user_vector;
70
71 fprintf(' v  =');
72 disp(s_multi);

```

10. Write a MATLAB code that prompts the user to enter two vectors, then displays the addition and the inner product of them, and the angle value between them.

Answer.

```

1  clc;
2
3  % Prompt the user to enter two vectors
4
5  f_vector = input('Enter the first vector a = ');
6  s_vector = input('Enter the second vector b = ');
7
8  if length(f_vector) == length(s_vector)
9
10     n = length(f_vector);
11
12     %-----
13     %-----ADDITION-----
14
15     %Initialize an element-wise addition vector to zero
16
17     c = zeros(1, n);
18
19     % Loop through each element of the two vectors to compute the addition
20
21     for i = 1:n
22         \hspace*{1cm} c(i) = f_vector(i) + s_vector(i);
23     end
24
25     %%% Another way:
26     %%% c = f_vector + s_vector;
27
28     fprintf('\n')
29     fprintf('a + b = ');
30     disp(c);
31
32     %-----
33     %-----INNER PRODUCT-----
34
35     %Initialize the inner product to zero
36
37     inner_product = 0;
38
39     %Loop through each element of the two vectors to compute the inner product
40
41     for i = 1:n
42         inner_product = inner_product + f_vector(i) * s_vector(i);
43     end
44
45     %%% Another ways:
46     %%% inner_product = sum(f_vector .* s_vector);
47     %%% dot(f_vector, s_vector),

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```

48     fprintf('a.b = %.2f\n\n', inner_product);
49
50
51     %-----
52     %-----Angle Value-----
53
54     %Check if either vector is a zero vector
55
56     if norm(f_vector) == 0 || norm(s_vector) == 0
57         disp('One of the vectors is zero, so it is not possible to find the angle value.
58         ');
59     else
60         %Compute the angle value between the two vectors
61
62         angle_value = rad2deg(acos(inner_product / (norm(f_vector) * norm(s_vector))));
63
64         fprintf('    = %.2f\n', angle_value);
65     end
66
67 else
68     disp('ERROR, these vectors are not in the same vector spaces.');
```

```

69 end
```